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Exploring the role of education on environmental quality and renewable energy: Do education levels really matter?

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ABSTRACT

The current study examined the impact of different levels of education on environmental quality and renewable energy. Panel data was drawn from the World Bank and OECD databases for 20 OECD countries for the period 1997–2019. Rigorous analytic procedures were employed to better address the methodological issues of existing literature for the analysis of panel data. In this regard, cross-section dependency and heterogeneous structure of the panel data were considered. Accordingly, Gengenbach, Urbain, and Westerlund's panel cointegration test was employed to examine whether there was a long-term relationship between the variables. Fully Modified Ordinary Least Square and Mean Group Dynamic Least Squares were employed to test the robust impact of independent variables on the dependent variable in the long run. Lastly, Dumitrescu & Hurlin's Granger non-causality test was conducted to determine the direction of the relationship between variables. Findings showed that while lower-level education and per capita GDP significantly increased CO_2 emissions, higher education significantly decreased it. The CO_2 emissions and per capita GDP posed significant negative influences on renewable energy. On the contrary, while both levels of education had positive influences, higher education appeared to be the most influential variable on renewable energy. The causality analyses yielded bidirectional causality relationships between CO_2 emissions and renewable energy, low-level education, and higher education; conversely, unidirectional causality relationships between CO_2 emissions and per capita GDP. Collectively, the study highlighted the significance of a highly educated workforce for promoting and creating a sustained environment by reducing CO_2 emissions and increasing renewable energy utilization.

1. Introduction

Environmental sustainability, which can be defined as the ability to protect the attributions valued in the physical world, is one of the collective responsibilities to ensure a sustainable future for the next generations (Sutton, 2004). Today's societies are genuinely concerned about irreversibly degrading adverse effects of climate change on the environment and failing to leave sustainable environments for future generations. Environmental degradation has emerged as a serious global problem plaguing both developing and developed nations. It gradually has attracted global attention as it not only triggers the global warming but also disrupts the global carbon cycle (Adebayo and Kirikkaleli, 2021; Bölük and Mert, 2014). Greenhouse gas accumulation, particularly carbon dioxide, is one of the most important contributing factors to global warming (Mohsin et al., 2021). The dual tasks of responding to the increasing energy needs toward maintaining reasonable living standards while supporting a dynamic economy continue to trigger environmental degradation and climate change (Bull, 2001). Developing policies to prevent environmental degradation and meeting the increased energy demands have become the most daunting duties of policymakers.

There is a widespread agreement about carbon dioxide (CO_2) emissions as being one of the main reasons for environmental degradation and it is used as an environmental quality indicator by the international community (Li et al., 2021a, 2021b; Mikayilov et al., 2018; Rehman et al., 2021). Economic growth and renewable energy are identified as the leading contributors of CO_2 emissions in the relevant literature. A growing number of studies also suggests that renewable energy systems reduce CO_2 emissions (Dogan and Seker, 2016; Özbuğday and Erbas, 2015; Piłatowska et al., 2020; Shafiei and Salim, 2014; Waheed et al.,

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2018). The economic growth concept stands out as a contributing factor to environmental change in the literature. A common finding of these studies pointed out the negative effects of economic growth on the environment and identified economic growth as a trigger of CO_2 emissions (Mikayilov et al., 2018; Mohsin et al., 2021; Zafar et al., 2021; Nathaniel et al., 2021; Pata and Yurtkuran, 2018; Özbay and Pehlivan, 2021; Danish Hassan et al., 2019). These studies further pointed out the bidirectional causality relationships between CO_2 and renewable energy (RE) (Dogan and Seker, 2016; Inglesi-Lotz and Dogan, 2018).

Poor environmental quality and global warming are considered one of the most severe problems in today's civilized world. The international community expressed a widespread desire to prevent global warming and protect the environment through the Paris Agreement in 2015. The Paris Agreement provides a binding legal framework and allowed international collaboration through capacity building, and financial and technical support (United Nations, 2015). Local policymakers also have enacted national policies and imposed substantial sanctions against the violators of policies to curb the magnitude of the insinuating problems. These policies also call for capacity building and raising public awareness through educational programs.

Another commonly accepted factor in fighting against environmental climate change and environmental degradation is the use of renewable energy. The OECD (2021) defines renewable energy as a type of energy that is extracted from hydro (except pumped storage), geothermal, solar, wind, tidal, and wave sources. According to an International Energy Agency (IEA) report (2020), several promising developments have taken place in this field that may potentially help the preservation of environmental quality. For example, in 2020, the annual renewable capacity increased by 45% to almost 280 GW, the highest increase since 1999 (IEA, 2021). The increase in renewable energy means less fossil-based energy, which in turn, less CO_2 emissions. Given the importance of renewable energy in discussions about reliable and sustainable energy production for the future, it is critical to understand its key determinants and draw implications for energy policy for creating a sustainable environment (da Silva et al., 2018).

Despite the positive developments, a considerable number of public energy policies worldwide continue to encourage the use of fossil energy, thus, hindering renewable energy investments (Aguirre and Ibikunle, 2014). Developing renewable energy policies is a necessity for ensuring a sustainable environment and future. As Bourcet (2020) pointed out in his systematic review of relevant literature, variables that are linked to economic development and energy consumption directly influence renewable energy initiatives. Yet, to the best knowledge of the authors, no study has been conducted about whether different levels of education would make any difference in the use of renewable energy. Also, Mahalik et al. (2021) stated that it is necessary to examine the effect of different types of education on the preferred energy utilization.

O'Neill et al. (2020) point out the links between education and climate change risks, which may include both synergies and trade-offs. Establishing a fine balance between these counter-effects remains uncertain. Even if the balance between these competing effects remains to be resolved, education has emerged as a crucial factor in eradicating the factors that negatively influence environmental quality (Marra and Colantonio, 2020). There is ample literature supporting this view. For instance, Kandpal and Broman (2014) identified education as one of the most effective ways of handling the problems that societies endure. Similarly, Tiba and Belaid (2021) noted that a better-educated society would help reduce environmental harm by enabling social sustainability. Moreover, education would play a significant role in developing social responsibility, which might in turn help decreasing CO_2 emissions (Alkhateeb et al., 2020). While the literature has consistently acknowledged its significance in combating CO_2 emissions, using education to fight against environmental degradation has been neglected in many countries (Eyuboglu and Uzar, 2021).

To make things even worse, the influence of different levels of education on environmental quality has not been adequately studied in the relevant literature. Existing literature supports the role of education in leaving a sustainable and quality environment for future generations. The significance of education in environmental quality stems from its function in raising the awareness of people on energy conservation and environmental protection and preservation. Educated individuals happen to be more sensitive to climate change as well as other social and environmental issues (Keser et al., 2010). Accordingly, increasing access to higher levels of education for all citizens should be a sustainable development priority for all nations across the world. To address these gaps in the relevant literature, the current study will examine the effects of lower- and higher education on environmental quality (i.e., CO_2 emissions) and renewable energy.

2. Theoretical foundations

A growing number of studies are conducted on the contributing factors of climate change and initiatives to alleviate its effects. Earlier studies mainly focused on the relationships between economic growth, renewable energy, and CO_2 emissions. These studies identified the human factor as both the cause of and the remedy for climate change and environmental problems. Some of these studies focused on human capital as an important ingredient of environmentally sensitive economic development. For instance, Sarkodie et al. (2020) examined the determinants of climate change and its detrimental effects in China. The authors concluded that over-reliance on fossil fuel and natural sources for the sake of economic growth, carbon-intensive trade, and carbon-embedded human capital hinders the efforts to lessen climate change and its adverse effects.

In their study in South Africa, Iorember et al. (2021) found that renewable energy, human capital, and increased trade improve the environmental quality by reducing the ecological footprint, while the increase in income encourages the ecological footprint. Moreover, they also suggested that economic growth causes an ecological footprint in the short run. Other studies examined different country contexts and provided valuable findings regarding the role of human capital in enhancing the use of renewable energy and curbing CO_2 emissions. For instance, studies by Mahmood et al. (2019); Ahmed et al. (2020); Hao et al. (2021); Zafar et al. (2019); Zhang et al. (2021); and, Pata and Caglar (2021) also produced similar results. Yao et al. (2020) underlined the role of improving human capital in decreasing CO_2 emissions over different geographical contexts. The second important finding in the relevant literature suggested that carbon-embedded human capital does not play a role in improving the environment; on the contrary, it harms the environment. This finding suggests that human capital is either insufficient in reducing environmental degradation or it contributes to environmental degradation.

The study by Eyuboglu and Uzar (2021), which employed the ARDL approach, illuminated the potential mechanisms between higher education and CO_2 emissions. This particular study concluded that the increase in higher education levels negatively influenced CO_2 emissions levels. Further, the study suggested that higher education may have different effects on CO_2 emissions in the long and short terms.

In this vein, by using the ARDL method, Alkhateeb et al. (2020) studied the effects of education, energy consumption, and economic growth on CO_2 emissions in Saudi Arabia. The authors found that while primary education did not have any significant effect on CO_2 emissions, the secondary education had a negative effect on CO_2 emissions in the Saudi context. Their study also showed a positive association between energy consumption and CO_2 emissions. In addition to the above findings, Mahalik et al. (2021) found that total energy consumption, economic growth and globalization, primary education, and non-renewable energy consumption contributed to increasing carbon emissions in the BRICS countries. As part of the environmentally sensitive development policies, Tsaurai and Ngcobo (2020)suggested investments in education in BRICS countries where renewable energy is considered as an integral instrument of economic development. According to these authors, the

findings their study showed that education reduced the size of the negative effect of renewable energy consumption on economic growth in the BRICS countries.

Czarnek et al. (2021) investigated the relationships between ideology, education, and climate change in 64 countries. Their study showed that education positively affects pro-climate change beliefs at low and moderate levels of development. Also, Tang et al. (2021) analyzed the relationships between environment and education in 114 countries. Their findings further supported the role of education in the use of renewable energy, thus, leading to diminishing environmental degradation. Özçiçek and Ağpak (2017) also explored the role of education level on non-hydro renewable energy consumption in 62 countries from the period of 1990 to 2014. Their findings found a positive influence of education on citizens' renewable energy participation levels. Moreover, the effect of higher education was found to be stronger than that of lower levels of education.

Although the role of education on renewable energy and environmental quality has been explored in these studies, the current literature failed to employ a comparative perspective on the effects of lower and higher levels of education. The existing literature also seems to lack studies that examined the effects of different levels of education on the utilization of renewable energy resources. Furthermore, the current studies also appear to have some methodological limitations. These studies appear to neglect the critical assumption of heterogeneity of panel data. The current study attempted to addressed these gaps in the relevant literature.

3. Methodology

3.1. Research design

The current study employed a longitudinal panel data research design. Panel data models follow a given sample of individuals over time and provide multiple observations on each individual in the sample (Hsiao, 2003). Panel data is the pooling of observations on a cross-section of countries, households, etc., across several time periods. Macro-panels usually involve a number of countries over 20 to 60 years (Baltagi, 2021). Panel models offer many advantages in bringing together cross-sectional observations over a period of time. The most crucial advantage of panel data analysis is that it allows the researcher great flexibility in modeling relationships by using large datasets (Özbay and Oğuztürk, 2020a).

The longitudinal design allowed the study to examine relationships between study variables over a period of time. In this regard, the study covered selected 20 OECD countries 23 years period from 1997 through 2019. The relationships between study variables were examined and compared without changing the existing independent variables such as education levels, GDP, etc.

3.2. Data sources

The sources of data for CO_2 and GDP was from the datasets of the *World Development Indicators*. The renewable energy (RE), lower level education (LE), and higher education (HE) data were collected from different data sets of the OECD. The panel data included the OECD countries (Australia, Canada, Costa Rica, Czech Republic, Estonia, France, Germany, Greece, Hungary, South Korea, Latvia, Mexico, Poland, Slovak Republic, Spain, Sweden, Switzerland, Turkey, United Kingdom, and the United States) spanning 23 years from 1997 through 2019. While all OECD member countries were targeted, only 20 countries were included in the analyses due to the lack of complete data for study variables for some member countries. Eighteen countries where the education variable could not be fully observed were also excluded from the study.

Parallel to the common practice in the relevant literature, CO_2 emission was used as the environmental quality measure in this study (Li et al., 2021a, 2021b; Mikayilov et al., 2018; Rehman et al., 2021). The RE represents renewable resources such as geothermal, solar, wind, tides, and waves. This indicator is also one of the most important variables in increasing environmental quality. According to Bourcet (2020), 58% of the studies take renewable energy supply as an indicator in the literature where renewable energy determinants are discussed. The Per capita GDP was used as a proxy for economic growth. The economic growth concept stands out in studies investigating environmental quality (Mikayilov et al., 2018; Mahalik et al., 2021; Zafar et al., 2021; Nathaniel et al., 2021; and Özbay and Pehlivan, 2021). Per capita GDP was commonly used in the literature as a determinant of RE (Bourcet, 2020). HE and LE represented the levels of education.

3.3. Specification of models

Two models were developed to test the hypothesized relationships between the study variables. The first model included proxies of environmental quality. The CO_2 emissions were treated as an endogenous variable that represented the environmental quality. The second model was created in order to present a comparative analysis with the first model by examining the effect of different levels of education, GDP and CO_2 on renewable energy, and to put forward more robust policy propositions in this context. The two models are shown in Eq. (1) and Eq. (2):

Model 1:

$$lnCO_{2,it} = \beta_0 + \beta_1 \mathrm{lnGDP}_{it} + \beta_2 lnRE_{it} + \beta_3 lnLE_{it} + \beta_4 \mathrm{lnHE}_{it} + \varepsilon_{it} \quad (1)$$

$i = 1,2,3,4\ldots.0.20.$
$t = 1,2,3,4\ldots0.23.$

Model 2:

$$lnRE_{it} = \beta_0 + \beta_1 \mathrm{lnGDP}_{it} + \beta_2 lnCO_{2,it} + \beta_3 lnBUE_{it} + \beta_4 \mathrm{lnTE}_{it} + \varepsilon_{it} \quad (2)$$

$\mathrm{i} = 1,2,3,4\ldots.0.20.$
$\mathrm{t} = 1,2,3,4\ldots0.23.$

3.4. Panel data assumptions tests

It is critical to screen the data for potential assumption violations for underlying multivariate techniques (Chun Won et al., 2017). These tests help to determine the method to be used in the study. The current study employed several data assumptions test strategies before data analyses. First, the assumption of normality of the variables was examined. The data were also examined for the multiple uniqueness and multicollinearity problems. The presence of a possible high correlation coefficient between the study variables may yield inaccurate results. Furthermore, the panel data was tested for cross-sectional dependence and heterogeneous assumptions. This allowed identifying appropriate analysis techniques for the data. Lastly, the model should be investigated whether it included unit-root or not. To determine the long-term relationship between variables, cointegration tests are used. To select the appropriate cointegration test, the stationarity degrees of the variables must be determined. On the stationarity degrees of determination of the variables, the unit root test is conducted.

3.4.1. Tests for skewness and kurtosis

To examine whether the data conformed to the normality assumption, the Skewness and Kurtosis tests were employed. The presence of asymmetric or heavy-tailed errors and non-normality could affect the statistical properties of inferential strategies. The Skewness and Kurtosis tests are based on Galvao et al. (2013). They can be seen as an extension of the classic Bera-Jarque test of normality for the panel data case (Alejo et al., 2015). These tests look for skewness and excessive kurtosis of data and allow researchers to identify divergences from gaussian, individually or jointly, in both error components of a standard panel regression. The bootstrapping method was utilized for the skewness and Kurtosis

tests. Experiments show that for sample sizes similar to those used in empirical practice, the practical dimensions are close to nominal values and the tests have outstanding strength characteristics (Galvao et al., 2013).

3.4.2. Correlation matrix and VIF statistics

The data were also checked for possible multicollinearity issues. Two methods are frequently used for detecting multicollinearity. The first of these two methods is the correlation matrix, and the second is Variance Inflation Factor (VIF) statistics.

The correlation matrix was obtained with the following Eq. (3):

$$r_{x,y} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n\sum x_i^2 - (\sum x_i)^2}\sqrt{n\sum y_i^2 - (\sum y_i)^2}} \tag{3}$$

$r_{x,\ y}$ is the correlation coefficient of the linear relationship between the x and y variables. The denominator gives uncorrected sample standard deviations for X and Y. The correlation coefficients should be in the range of −1 and + 1. However, it should be smaller than the difference between −1 and + 1 (Francis et al., 1999). The preferred range is usually between 0.8 and − 0.8 (Katz, 2011; Özbay, 2020b). Correlations beyond these ranges may cause multicollinearity or singularity problems.

Multicollinearity may also indicate the presence of an unusual variance of coefficients. VIF statistic is calculated as follows in the Eq. (4):

$$VIF_i = \frac{1}{1 - R^2} \tag{4}$$

The VIF value must be lower than 10, which means the effect of multicollinearity among variables does not exist (Natsagdorj et al., 2021; Patil and Sahu, 2016).

3.4.3. Cross-section dependence and slope heterogeneity test

Cross-section dependence and homogeneity assumptions should be examined to ensure that the results are not biased and the right tests are selected. Pesaran's (2004) test of cross-section dependence was employed to determine cross-section dependence in the study. This test was particularly appropriate as the panel data was non-stationary and included structural breaks and heterogenous samples. Pesaran (2004) developed the following equation for the unbalanced panels which are based on pair-wise correlation coefficients Eq. (5):

$$CD = \sqrt{\frac{T}{N(N-1)}}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N-1}\sqrt{T_{ij}}\,\acute{\rho}_{ij} \tag{6}$$

T_{ij} The correlation number refers to the number of the calculated observations. Under the H_0 hypothesis where there is no inter-unit correlation, this statistic has a normal distribution if $T_{ij} > 3$ and N is large enough.

Swamy's (1970) and the Pesaran and Yamagata's (2008) slope heterogeneity test was used to determine whether the model was homogeneous. To test Swamy's (1970) random coefficient regression, one should look at the difference between the unit-specific OLS estimators that ignore the panel structure of the data and the weight-mean matrices of WE. If there is no statistical difference, the parameters are homogeneous (Tatoğlu, 2018). Accordingly, the hypothesis to be tested was:

H_0=$\beta_i = \beta$.

The hypothesis calls for the parameters to be homogeneous. Pesaran and Yamagata (2008) suggest two different test statistics for slope heterogeneity tests in large and small samples. $\widetilde{\Delta}$, refers to the slope heterogeneity test.

$$\widetilde{\Delta} = \sqrt{N}\left(\frac{N^{-1}\widetilde{S} - k}{\sqrt{2k}}\right) \tag{7}$$

Pesaran and Yamagata's (2008) developed the small sample properties of the distribution tests under the normal distribution of errors, considering the following mean and variance bias-adjusted versions:

$$\widetilde{\Delta}adj = \sqrt{N}\left(\frac{N^{-1}\widetilde{S} - E(\widetilde{Z}_{iT})}{\sqrt{var(\widetilde{Z}_{iT})}}\right) \tag{8}$$

3.5. Unit-root tests

It is imperative to determine whether the variables are stationary in level or difference. Determining the appropriate method according to the unit-root test results prevents biased results. Unit-root tests can be divided into as first-generation and second-generation, based on cross-sectional dependence. In this study, both first and second-generation tests are used.

3.5.1. Im, Pesaran and Shin (IPS) panel unit-root test

The IPS test, which allows errors to be autocorrelated and when T and N are large enough, can be expressed as follows (Im et al., 2003):

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + \sum_{j}^{\rho_i}\rho_{ij}\Delta y_{i,t-1} + \varepsilon_{it}, i = 1....N; t = 1.....T \tag{9}$$

ρ_i the persistence parameter is assumed to change among countries. Inference about stationary data is made based on the *p*-values of the results. When t represents time series, $t = 1, 2, \ldots 23$.

Here Δy_{it}, $\Delta y_{i,\ t}$is for the specific time period and i is the difference among countries.

This inference is made under two hypotheses.

H_0=Data are not stable

H_1=Data are stable

When we allow for serial correlation and heterogeneity during the basic data generation process, the simulation results yielded strong results if a sufficiently large delay order is chosen for the ADF regressions which are taken as the basis (Im et al., 2003). As a result, to reduce the correlation between units, the IPS unit-root test, which allows the autoregressive parameter to be heterogeneous, can be applied to the series whose cross-sectional means are calculated (Tatoğlu, 2018).

3.5.2. Pesaran CIPS unit-root test

Pesaran (2007) bases the unit-root hypothesis on the t-rate of unit-root tests for single-factor removed models with the following cross-sectionally enhanced non-correlation errors:

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + c_{ij}\overline{y}_{i,t-1} + d_i\Delta y_i + \varepsilon_{it}, i = 1....N; t = 1.....T \tag{11}$$

The CADF regression, which takes serial correlation in error terms or factors, was estimated in the following form:

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + c_{ij}\overline{y}_{i,t-1} + \sum_{j=0}^{\rho_i} d_{ij}\Delta\overline{y}_{i,t-j} + \sum_{j=1}^{\rho_i}\delta_{ij}\Delta\overline{y}_{i,t-j} + \varepsilon_{it}, i = 1....N; t = 1.....T \tag{12}$$

Where α_i*is the* country-specific affect, $\delta_{ij}$is deterministic time trend, ε_{it} and denotes estimated residuals. Pesaran (2007) stated that the above form is not a first-order error process, but the approach would easily be extended to higher-order processes. For instance, the relevant individual CADF statistics for an AR (p) error specification revealed above with the OLS t-ratio of β_i in the path order cross-section/time series incremental regression. As a result, Pesaran (2007) provides a powerful estimator to test unit roots in dynamic panels that are subject to cross-sectional dependent and series-related errors.

This inference is made under two hypotheses.

H_0=$\beta_i = 0$ (for all i)

against the possibly heterogeneous alternatives,

H_1= $\beta_i < 0$ ($i = 1,2, \ldots..N_i$, $\beta_i = 0$, i=N_i+1, N_i+2, …..N)

As a result of the unit-root test applied to the models, the cointegration relationship between the stationary models in difference should

be examined.

3.6. Data analysis strategy

This section presents the data analysis techniques employed to address the study hypotheses.

3.6.1. Gengenbach, Urbain, and Westerlund panel Cointegration tests

This test examines the long-term relationship between variables. When the variables are not stationary at the level, this would cause spurious regression relations. Suppose the series is not stationary at the level (I(1)) and is cointegrated. In this case, there is a long-term relationship between these series, and there is no spurious regression in this relationship.Gengenbach et al. (2016) developed new cointegration tests predicated on the importance of the EC term in a panel EC model with nonstationary common factors. Based on this EC (error correction) model, the test in the vector format is written as follows:

$$\Delta y_i = d\delta_{y.x_i} + a_{y_{i,-1}} + \omega_{i,-1} y_i + \upsilon_i \pi_i + \varepsilon_{y.x_i} = a_{y_i} y_{i,-1} + g_i^d \lambda_i + \varepsilon_{y.x_i} \tag{13}$$

The panel test statistic, which takes cross-sectional dependence according to the units into account and uses the cross-sectional averages, is as follows:

$$T_C = \frac{1}{N} \sum_{i=1}^{N} T_{C_i} \tag{14}$$

which tests $H_0 = a_{y1} = a_{yN} = 0$ versus
$H_0 = a_{y1} < 0$ for at least some i.

3.6.2. Fully modified ordinary least squares (FMOLS) and mean group dynamic least squares (DOLSMG)

If a long-term cointegrating relationship is found between the series, the relationship of the models created is determined with the help of the error correction model. This study employed two methods to determine whether or not there were deviations from the assumption. While determining these methods, this study took the cross-section dependence of the model and whether it has a heterogeneous structure into consideration. Pedroni (1996, 2000) stated when properly configured to measure the potential heterogeneity and fixed effects in the intrinsic dynamics associated with such panels, asymptotic distributions for these estimators will be centered around the true value and free of disturbing parameters. He also stated that this test allows heterogeneity in cross-sections.

The heterogeneity panel model is generally

$$y_{it} = \alpha_i + \beta x_{it} + u_{it} \tag{15}$$

$$x_{it} = x_{it-1} + \varepsilon_{it} \tag{16}$$

When defined, the FMOLS estimator is obtained in this way.

$$\widehat{\beta}^*_{NT} - \beta = \left(\sum_{i=1}^{N} \widehat{L}_{\varepsilon i}^{-2} + \sum_{t=1}^{T} (x_{it} - \overline{x}_i)^2 \right)^{-1} \sum_{i=1}^{N} \widehat{L}_{ui}^{-1} \widehat{L}_{\varepsilon i}^{-1} \left(\sum_{t=1}^{T} (x_{it} - \overline{x}_i) u^*_{it} - T\widehat{y}_i \right) \tag{17}$$

$\widehat{L}_i$ is a lower triangular decomposition. Error terms are stable and y_{it} has a first-order integration relationship, it can be said that there is a long-term cointegration between y_{it}andx_{it}. FMOLS t statistic is obtained as follows.

$$t_{\widehat{\beta}^*_{NT}} = \left(\widehat{\beta}^*_{NT} - \beta \right) = \left(\sum_{i=1}^{N} \widehat{L}_{\varepsilon i}^{-2} + \sum_{t=1}^{T} (x_{it} - \overline{x}_i)^2 \right)^{1/2} \rightarrow N(0,1) \tag{18}$$

This model is resistant to autocorrelation (Pedroni, 2000). However, there is no effective predictor in the case of cross-section dependence. Therefore, before interpreting the results of the model, a cross-section dependency test should be performed on the residues. To control the internality of the DOLSMG estimator and the DOLS regression estimator, the leading and lag differences of the regressors were constructed:

$$y_{it} = \alpha_i + \beta_i x_{it} + \sum_{k=-K_i}^{K_i} \gamma_i \Delta x_{it-k} + u^*_{it}, \tag{19}$$

From this regression, the DOLSMG panel estimator is constructed in this way:

$$\widehat{\beta}^*_{DMG} = \left[N^{-1} \sum_{i=1}^{N} \left(\sum_{t=1}^{T} Z'_{it} Z_{it} \right)^{-1} \left(\sum_{t=1}^{T} Z'_{it} \widetilde{Y}_{it} \right) \right]_1 \tag{20}$$

Here $Z_{it}$and $\widetilde{Y}_{it}$ terms respectively
Z_{it} is 2(K + 1) X 1 vector of regressors (explanatory variables)

$$Z_{it} = (x_{it} - \overline{x}_i, \Delta x_{it-K}, \ldots \Delta x_{it+K}),$$

$$\widetilde{Y}_{it} = y_{it} - \overline{y}_i,$$

A subscript of 1 outside the brackets indicates that only the first element of the vector is taken to obtain the pooled slope coefficient.

DOLSMG t statistic is obtained as follows:

$$t_{\widehat{\beta}^*_{DMG}} = N^{-1/2} \sum_{i=1}^{N} t_{\widehat{\beta}^*_{Dols,i}} \tag{21}$$

The averages of DOLS estimators for each "i" were considered in the analyses.

3.6.3. Dumitrescu & Hurlin granger non-causality test

In panel data analysis, different techniques are used to determine the direction of the relationship between variables for each cross-section and in general. Dumitrescu and Hurlin (2012) developed a causality test that is insensitive to the differences in size between the time dimension and the cross-section size. This technique produces effective results even when the time dimension is smaller than the cross-section size. Another main advantage of this test is that it can also be applied to unbalanced panels. Moreover, it also allows studying cross-sectional dependence and heterogeneity (Dumitrescu and Hurlin, 2012). For each individual $i = 1,..,N$, at time $t = 1,..,T$, this test considers the following linear heterogeneous model:

$$y_{it} = \alpha_i + \sum_{i=1}^{K} \gamma_i^{(k)} y_{i,t-k} + \sum_{i=1}^{K} \beta_i^{(k)} x_{i,t-k} + \varepsilon_{it}, \tag{22}$$

where $K \in N^*$ and $k \in N^+$, $\beta_i = (\beta_i^{(1)}, \ldots, \beta_i^{(k)})'$. α_i, $\gamma_i^{(k)}$ and $\beta_i^{(k)}$ indicate constant term, lag parameter, and coefficient slope, respectively.

4. Findings

The purpose of the study was to comparatively examine the influence of different levels of education on environmental quality and renewable energy utilization. The innovative data analytic strategy not only better addressed the panel data but also helped investigate the directions of the relationship between the study variables. To achieve the purposes of the study, it was essential to determine the appropriate methodologies. The results of the descriptive statistics and assumption tests are presented in the first section. The results of the two models are shown in the second section.

4.1. Descriptive statistics and assumption tests

Table 1 demonstrates the study variables and their summary statistics. The standard deviations showed that there was no volatility in the variables. Similarly, the test for normality showed that the data were normally distributed. When original continuous data does not follow the bell curve, it can take the logarithm in this data to make it as "normal" as possible. The statistical analysis results from this data become more

Table 1
Descriptive statistics and test for normality.

Variable	Classification	Mean	Std. Dev.	Min	Max	Observations
lnRE	overall	2.168729	0.9700677	−0.9162908	3.969914	$N = 459$
	between		0.9187398	−0.2949691	3.867227	$n = 20$
	within		0.3767444	1.008149	3.649611	$T = 22.95$
$lnCO_2$	overall	1.912481	0.6202038	0.0953102	3.010621	N = 459
	between		0.6278491	0.3046397	2.862827	n = 20
	within		0.1190166	1.501928	2.21058	T = 22.95
lnLE	overall	3.095628	0.6743138	1.808289	4.369448	$N = 460$
	between		0.6499951	2.230182	4.236203	n = 20
	within		0.2290322	2.490023	3.743465	$T = 23$
lnHE	overall	3.229087	0.4288526	2.014903	4.084294	N = 460
	between		0.3755374	2.51274	3.870657	n = 20
	within		0.2228133	2.610761	3.80739	T = 23
lnGDP	overall	10.04093	0.6954031	8.647655	11.28017	N = 460
	between		0.6923203	8.932305	11.18493	n = 20
	within		0.1649431	9.437863	10.45678	T = 23

Joint test for Normality on e: chi2(2) = 0.12 Prob > chi2 = 0.9395
Joint test for Normality on u: chi2(2) = 3.95 Prob > chi2 = 0.1389

valid. The lnRE, $lnCO_2$, lnLE, lnHE, and lnGDP represent variables in natural logarithm format.

Table 2 shows the correlations and the VIF statistics. Correlation coefficients were expected to be between −0.8 and 0.8 and the VIF statistic is expected to be between 1 and 10. The VIF value was lower than 10, which means the effect of multicollinearity among variables does not exist (Natsagdorj et al., 2021; Patil and Sahu, 2016). The VIF values were lower than 10, which shows that the effect of multicollinearity among lnRE, $lnCO_2$, lnLE, lnHE, and lnGDP do not exist. The study variables were not significantly correlated; suggesting that there were no multicollinearity or singularity issues.

The unit-root tests depend on whether the variables are homogeneous and whether they contain Cross-Section Dependence (CD). If they do not have CD, the first generation tests should be preferred, and, if they do, the second-generation tests should be selected (Tatoğlu, 2018). Unit-root tests that take into account the variables' homogeneous structure and generation separation should be determined in the examinations. Thereby the authors conducted the Pesaran CD test to determine which generation unit-root test should be used. Below are the Pesaran CD test results.

As seen in Table 3, the variables have cross-sectional dependence. Unit-root tests and selected panel tests were applied depending on whether the model was homogeneous or not. Table 4 presents the findings of the test and the Pesaran and Yamagata (2008) homogeneity tests.

As shown in Table 4, the null hypothesis was rejected, thus, it was concluded that the model was heterogeneous. Table 5 demonstrates the results of first and second-generation unit-root tests to test the heterogeneity of the variables.

The findings of unit-root test results showed that the study variables were stable at the I(I) level. The unit-root test results suggested the investigation of the cointegration relationships between the study variables.

Table 2
Matrix of correlations.

Variables	(1)	(2)	(3)	(4)	(5)
(1) $lnCO_2$	1.000				
(2) lnGDP	0.525	1.000			
(3) lnRE	−0.513	−0.068	1.000		
(4) lnLE	−0.500	−0.338	−0.003	1.000	
(5) lnHE	0.430	0.714	0.085	−0.412	1.000

Mean VIF:1.66

Table 3
Pesaran CD test.

Variable	CD	P-val
lnRE	32.59	0.000
$lnCO_2$	15.39	0.000
lnGDP	56.59	0.000
lnLE	58.70	0.000
lnHE	62.09	0.000

Notes: Under the null hypothesis of cross-section independence CD ~ N (0,1).

Table 4
Testing for slope heterogeneity.

		Delta	p-value
Pesaran and Yamagata S Test		19.892	0.000
	adj.(Δ)	23.147	0.000
Swamy S Test		chi2(95) = 78,877.91	
		P prob.>chi2=0.0000	

4.2. Findings for testing the models

Gengenbach et al.'s (2016) cointegration test was employed as the established model was unstable and heterogeneous. This test also allows cross-sectional dependence.

Table 6 and Table 7 show that there was a cointegration relationship between the variables for Model one and Model two. Moreover, the inter-unit correlation test for the the model's residuals showed that Models one and two were resistant to cross-sectional dependence.

The model revealed the presence of cointegration relationships between the variables. The FMOLS and DOLSMG tests were conducted to estimate the long-run coefficients per the findings of the presence of a long-run relationship through the Gengenbach et al. (2016) panel cointegration test. Both of the tests can be used for heterogeneous panels. While FMOLS does not consider CD (Gür, 2018), DOLSMG is resistant to CD. Estimators that ignore the inter-unit correlation are referred to as first-generation estimators (Tatoğlu, 2018). Second-generation estimators were resistant to inter-unit correlation.

Table 8 shows the FMOLS and DOLSMG test results for model one. The presence of CD between units for the FMOLS estimators may yield possible biased results. Therefore, the results of DOLSMG should be included in the interpretations. When the time dummy variable for DOLSMG is deducted from the model, the coefficient of lnRE variable was −0.09143, t-stat −19.46. This finding suggested a possible long-term negative effect of RE. While lnLE (0.492) and lnGDP (1.004)

Table 5
Panel unit-root tests.

	CIPS				LM, PESERAN AND SHIN (IPS)			
	without trend	with trend	without trend First difference	with trend First difference	without trend	with trend	without trend First difference	with trend First difference
lnRE	−0.112	0.620	−5.955***	−6.139***	1.8674	1.1972	−16.8120***	−16.2275***
lnCo2	3.370	−0.468	−8.981***	−7.465 ***	3.2834	−2.4565**	−17.0343***	−15.7913***
lnGDP	−2.889***	0.278	−3.483 ***	−1.002	2.8553	−0.5327	−8.1894***	−5.8376***
lnLE	−0.029	−1.585	−7.230 ***	−5.571***	−0.0691	−1.3752	−15.7805***	−13.6164***
LnHE	−2.385***	−0.741	−8.338***	−7.205***	−0.0691	−1.3752	−15.7805 ***	−13.6164***

Notes: *** indicates significance at a 1% level. **indicates significance at.%5 level The null hypothesis is non-stationarity.

Table 6
Panel cointegration test for the first model.

d.y	Coef	T-bar	P-val*
y(t-1)	−1.219	−4.131	≤0.01

Pesaran (2015)**CD-test**		
Variable	CD	P-val
$lnCO_2$	14.950	0.000
lnRE	29.337	0.000
lnGDP	55.465	0.000
lnLE	55.958	0.000
lnHE	61.964	0.000
e	0.528	0.598

Table 7
Panel Cointegration Test for the Second Model.

d.y	Coef	T-bar	P-val*
y(t-1)	−1.009	−4.302	≤0.01

Pesaran (2015)**CD-test**		
Variable	CD	P-val
lnRE	29.337	0.000
$lnCO_2$	14.950	0.000
lnGDP	55.465	0.000
lnLE	55.958	0.000
lnHE	61.964	0.000
e	0.928	0.354

Root mean square error: 0.0107
Number of observations: 439
Number of groups: 20

Table 8
Long-term estimators for first model.

Variables	DOLSMG		FMOLS	
	Beta	t-stat	Beta	t-stat
lnRE_td	0.005***	−10.170	−0.27***	−70.77
lnGDP_td	1.004***	51.720	0.32***	29.97
lnLE_td	0.492***	17.890	0.17***	6.60
lnHE_td	−0.038***	9.551	0.09***	−11.16
			CD-test	p-value
			49.830	0.000

affected CO_2 emissions positively, lnGDP variable was the strongest predictor of CO_2 emissions. lnHE variable, on the other hand, had a negative significant effect (−0.038) on CO_2 emissions.

Table 9 presents the long-term relationships of the variables for the entire sample in model two. Beta represented the long-term parameter. Findings yielded a cross-sectional dependence between the study variables via the FMOLS test. Therefore, the DOLSMG estimator, which is resistant to cross-sectional dependence, was interpreted. The results also suggested that, for the whole model, $lnCO_2$ (−0.769) and lnGDP (−0.792) negatively influenced RE. Both the lnLE (0.223) and lnHE (1.413) variables had significant positive influences on RE. The lnHE variable was the strongest significant predictor of RE.

Table 9
Long-term estimators for the second model.

Variables	DOLSMG		FMOLS	
	Beta	t-stat	Beta	t-stat
$lnCO_2$	−0.769***	−6.061	−1.05***	−64.97
lnGDP	−0.792***	3.241	−0.16	−1.60
lnLE	0.223***	−3.851	−0.67***	−6.20
lnHE	1.413***	21.080	0.35 ***	27.41
			CD-test	4.800
			p-value	0.000

Dumitrescu and Hurlin's (2012) Granger non-causality test yields strong results for cross-section dependence and heterogeneity models. The observations for the causality test must be without gaps (no missing values allowed). For this reason, accordingly, Costa Rica, with one missing observation for $lnCO_2$ and lnRE, was excluded from the model just for the causality test. The study determined the delays (AIC) in the model by the information criterion.

As seen in Table 10, the causality analysis revealed the presence of a bidirectional relationship between CO_2 emissions and renewable energy (RE), lower level education, and higher education. Also, there was a unidirectional causality from CO_2 emissions to per capita GDP. These results also confirmed the long-term regression coefficients and showed that education level is one of the most important factors for environmental quality and renewable energy. In addition to the above results, the fact that there is a bidirectional causal relationship between environmental quality, renewable energy, and education which suggests that environmental quality and renewable energy affect education.

Table 10
Dumitrescu & Hurlin granger non-causality test results.

Direction of Causality	W-bar	*Z*-bar	p-value
$lnCO_2$→ lnRE	**3.191**	**6.7536**	**0.000*****
lnRE→$lnCO_2$	**5.754**	**14.652**	**0.000*****
$lnCO_2$ →lnGDP	**5.7066**	**14.5068**	**0.000*****
lnGDP →$lnCO_2$	2.6065	1.3219	0.1862
lnCo2→lnLE	**5.6211**	**14.2431**	**0.000*****
lnLE→$lnCO_2$	**9.8454**	**6.6789**	**0.000*****
$lnCO_2$→lnTE	**7.0236**	**18.5658**	**0.000*****
lnHE→$lnCO_2$	**2.1778**	**3.6301**	**0.000*****
lnRE→lnGDP	**11.9868**	**9.6307**	**0.000*****
lnGDP→lnRE	**12.3374**	**10.1140**	**0.000*****
lnRE→lnLE	**11.4991**	**8.9583**	**0.000*****
lnLE→lnRE	**14.5030**	**13.0990**	**0.000*****
lnRE→lnHE	**3.8678**	**8.8391**	**0.000*****
lnHE→lnRE	**9.7160**	**6.5005**	**0.000*****

The *** symbol indicates that the variables are significant at the 1% level.

5. Conclusions, implications, and recommendations

The first purpose of the current study was to investigate the impact of different levels of education on the CO_2 emissions, where the CO_2 emission was treated as a proxy variable for environmental quality. The second purpose of the present study was to examine the contributing factors of renewable energy. The dataset included a panel of data selected for the 20 OECD countries from 1997 to 2019. The Gengenbach et al.'s (2016) panel cointegration, fully modified ordinary least squares, and Mean Group Dynamic Least Squares, and lastly, Dumitrescu and Hurlin's (2012) Granger non-causality tests were employed for the analyses of panel data. The findings revealed causal relationships between environmental quality and per capita GDP, education levels, and RE.

The findings of the study showed the presence of a significant relationship between education and environmental quality. While higher education had a positive significant impact on environmental quality, lower level education had a negative significant influence on environmental quality. The findings of the study also showed that the relationship between CO_2 emissions and environmental quality differ by education level.

The relevant literature points out a negative relationship between education and CO_2 emissions; suggesting that improved educational opportunities improve environmental quality. Studies by Li et al. (2021a), Zafar et al. (2021), Hao et al. (2021), Mehmood (2021), and Zaman et al. (2021) reported similar results. The findings of the current study supported the common view in the literature on the impact of education on CO_2 emissions Mahalik et al. (2021). More specifically, higher education reduces CO_2 emissions and lower or basic education contributes to CO_2 emissions. However, the relevant literature mainly overlooked the differentiating impact of educational levels. Existing literature focused on the impact of higher education, human capital index, and educational expenditures on CO_2 emissions. The findings of this study also showed that RE contributes to a sustainable environment. Mohsin et al. (2021), Hao et al. (2021), and Mehmood (2021) noted similar findings.

There is momentum in studies examining the determinants of RE (Oğuztürk and Özbay, 2022; Bourcet, 2020; Liu et al., 2020; Bilan et al., 2019; da Silva et al., 2018; Apergis and Payne, 2014; Sadorsky, 2009). However, these studies are still limited. Also, to the best of the knowledge of the authors, whether citizens' education levels may have any effect on RE supply or the use RE has not been investigated yet.

The findings of the study showed that both low-level and high-level educated citizens contributed to a significant positive effect on RE. Moreover, the study's findings showed that the higher education level had the strongest impact on RE. Findings suggest that well-educated people have the greatest positive impact on RE. On the other hand, it was noticed that the rise in income affects RE negatively while it affects CO_2 emissions positively. With these findings, it can be concluded that higher education reduces environmental degradation for the whole panel while the increase in the number of people with basic education raises degradation. GDP per capita seems to be the most important factor for the increase in CO_2 emissions. Findings suggest that the level of education in developed countries plays a more important role in achieving a sustained environmental quality. In these nations, environmental degradation will decrease as the level of education increases.

The fact that the increase in the number of individuals with higher education has a positive effect on renewable energy supply does not mean anything on its own. It is essential that this increase also should have a negative effect on CO_2 emissions. The same result was not seen in people with basic education since basic-level education positively impacted CO_2 emissions. Moreover, non-renewable energy consumption and economic growth may also contribute to CO_2 emissions in the short run. These findings collectively further highlighted the role of increasing higher education opportunities to overcome environmental problems.

The growth concept stands out in studies investigating climate change. Negative effects of the economic growth on the environment were commonly mentioned as the triggers of CO_2 emissions. In addition to this view, Mikayilov et al. (2018), Mohsin et al. (2021), Zafar et al. (2021), Nathaniel et al. (2021), Danish Hassan et al. (2019), and Zafar et al. (2019) found similar results. These studies also suggested that as the level of income increases, environmental pollution and destruction increase at the beginning. However, there is a consensus in the literature about environmental improvements start only after higher income levels are maintained (Albayrak and Gökçe, 2015).

The findings of the study have both policy and practical implications. Policymakers who aim for creating sustainable environments need to promote policies that target improving educational opportunities for citizens' high levels of education. In this regard, increasing the higher education opportunities of disadvantaged populations has a particular policy importance. Policymakers and practitioners can contribute to these efforts by not only offering higher educational opportunities but also offering training opportunities that emphasize the importance of environmental sustainability and renewable energy consumption. Formal and informal training initiatives also may improve the sensitivity to environmental issues and would create awareness of such issues.

According to the OECD (2020), the main task for implementing national and international low-carbon strategies and addressing climate change is to further decouple GHG emissions from economic growth. The findings showed that while the effect of economic growth on CO_2 emissions were positive, the increased educational opportunities may combat or curtail its negative influences on the environment. This finding also revealed that the OECD countries as well as other countries with different developmental levels might not implement low carbon strategies. It may be concluded that the OECD countries have not yet achieved a low carbon strategy or an environmental economic growth policy that takes into account of effects of climate change. In other words, the energy factor that is the locomotive of economic growth is also the leading cause of high carbon emissions in these countries. The production entities supporting economic growth should urgently focus on environmentally friendly and renewable energy strategies. More specifically, OECD countries should further decouple GHG emissions from economic growth.

These findings also suggest the need for further research on the differing effects of education levels on environmental quality as well as the practical strategies to induce lower-level education for increased environmental awareness. Similarly, there is a need for further research on the effects of education levels in countries with different levels of economic development. Such research would illuminate whether the findings for the OECD and developed countries may apply to countries at different development levels. There is particularly a great need for studies examining the relationships between education, renewable energy, and environmental quality. It is hoped that the current study has provided an initial understanding of this neglected area of study and provided an impetus for further research.

In summary, the current study is one of the first studies that examine the effects of education levels on renewable energy consumption. The study examined the impact of education level on environmental quality, thus providing a comprehensive perspective by comparing the impact of education level on renewable energy. The findings of this study showed that the level of higher education not only improves environmental quality but also has the most significant impact on renewable energy utilization.

Another striking result of the study was that economic growth increases environmental degradation and had a negative effect on renewable energy utilization. This result indicated that countries included in this study did not have high levels of environmental sensitivity while they were economically growing. Above all, the current study and its findings highlighted a social policy issue by unveiling the effects of two different levels of education on the environment. The importance of higher education opportunities for the future has now been more clearly emphasized from an environmental point of view.

While the findings offered a clear understanding of the relationships

between educational levels, renewable energy, and climate, readers should apply caution to the generalization of the findings to other settings. Future studies should explore the relationships of study variables in different settings, such as isolated country-level analyses as well as country groups with different developmental levels. Examination of different contexts as well as different variables would help researchers, policymakers, and practitioners a better direction in creating and sustaining a livable world.

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Data sources and raw data

The data sources and raw data can be reached through the following link: https://data.mendeley.com/datasets/xdt5dpyx8c/1

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1. Variable definitions and data source.

Variable	Definition	Source
RE	It represents renewable energy. This variable is defined as the contribution of renewables to the total primary energy supply	OECD (2021), Renewable energy (indicator). doi: https://doi.org/10.1787/aac7c3f1-en (Accessed on 03 May 2021)
CO_2	CO_2 emissions (metric tons per capita). Carbon dioxide emissions are those produced by the combustion of fossil fuels and the production of cement. They include carbon dioxide produced during the consumption of solid, liquid, and gas fuels and gas flaring.	World Bank (2021), World Development Indicators: https://creativecommons.org/licenses/by-nc/4.0/ (Accessed on 03 May 2021)
GDP	is gross domestic product divided by midyear population… Data are in constant 2015 U.S. dollars. Gross domestic product per capita. It was the proxy for economic growth	World Bank (2021), World Development Indicators: https://datacatalog.worldbank.org/public-licenses#cc-by, (Accessed on 03 May 2021).
LE	Below upper education/lower level education. This variable is represented by people aged 25–64 who only had basic education (compulsory education).	OECD (2021), Adult education level (indicator). doi:https://doi.org/10.1787/36bce3fe-en (Accessed on 25 May 2021)
HE	Higher education: Tertiary Education. The tertiary education' variable is defined as individuals who have completed the highest level of education by the age group of 25–65.	OECD (2021), Adult education level (indicator). doi:https://doi.org/10.1787/36bce3fe-en (Accessed on 25 May 2021)

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.crsust.2022.100185.

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